Roll No. .....

TME–406

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

MECHANICS OF MATERIALS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Deduce expression to determine the elongation of a bar of trapezoidal section with uniform thickness. (CO1)

OR

(b) Determine the total compression of a bar made up of three circular cross-sections as shown in figure, the diameters being 10 mm, 20 mm and 30 mm of the top, middle and the bottom portions respectively. (CO1)

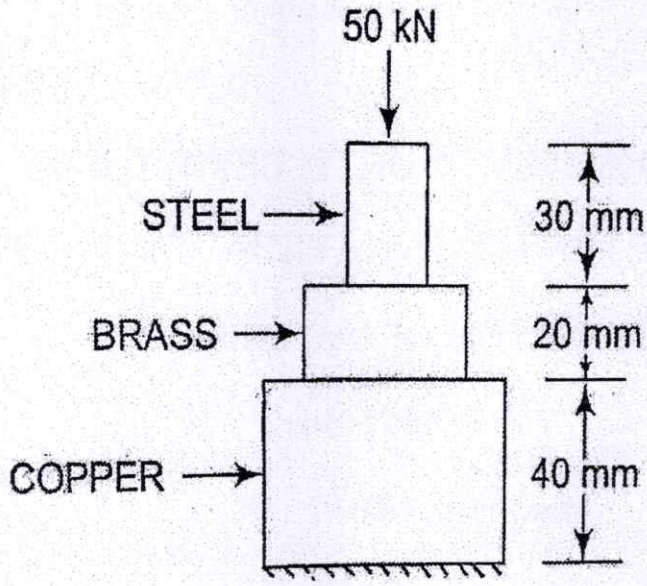
Take :

E(Steel) = 210 GPa

P. T. O.

E(Brass) = 105 GPa

E(Copper) = 100 GPa



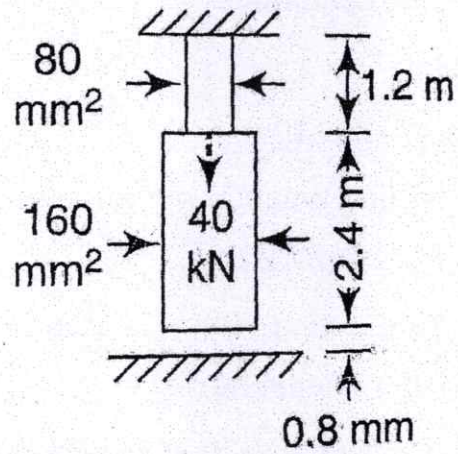
2. (a) (i) Deduce the expression for elongation of a bar of rectangular section due to self weight.

(ii) A steel wire of 8-mm diameter is used to lift a weight of 1.5 kN at its lowest end. The density of the wire material is 8000 kg/m^3. Determine the elongation of the wire if the length of the wire is 100 m. E = 205 GPa. (CO1)

OR

(b) A suspended bar system consists of two cross-sections as shown in figure. Initially its lower surface is 0.8 mm above the ground surface. Determine the reaction of the lower support and the stresses in each

section when a load of 40 kN is applied as shown in the figure. Take E = 205 GPa. (CO1)



3. (a) What do you mean by thermal stresses ? Explain. (CO2)

OR

(b) Rails are laid such that there is no stress in them at 24°C. If the rails are 32 m long. Determine :

(i) Stress in the rail at 80°C, when there is no allowance for expansion

(ii) Stress in the rail at 80°C, when there is expansion allowance of 8 mm per rail. (CO2)

(iii) The expansion allowance for no stress in the rail at 80°C.

Take :

Coefficient of linear expansion = 11 × 10^ (– 6)/°C, E = 205 GPa

4. (a) What do you mean by principal planes and principal stresses ? Derive the expression for principal stresses for a body subjected to direct and shear stresses. (CO2)

OR

(b) Draw Mohr's stress circle indicating the radius and centre point coordinate for given stress conditions. (CO2)

(i) Pure shear (60 MPa)

(ii) Hydrostatic stresses (120 MPa)

(iii) Normal stress in X-direction 120 MPa (tensile) and Normal stress in Y-direction 60 MPa (tensile)

(iv) Normal stress in X-direction 120 MPa (tensile) and Normal stress in Y-direction 60 MPa (compressive)

(v) Normal stress in X-direction 120 MPa (tensile) and Normal stress in Y-direction 60 MPa (tensile) with 60 MPa shear stress

5. (a) What is Mohr's stress circle ? How is the Mohr's circle constructed ? (CO2)

OR

(b) Two perpendicular tensile stresses of 300 MPa and 150 MPa act at a point in a material. Draw the Mohr's stress circle and find the normal and shear stress components on a plane the normal of which makes an angle of 40° with the 300 MPa stress. Also, find the resultant. (CO2)

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TME–405

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

KINEMATICS OF MACHINES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the types of kinematic pairs with examples in detail. (CO1, CO1)

OR

(b) What do you mean by degree of freedom of a kinematic pair ? Give examples. (CO1, CO1)

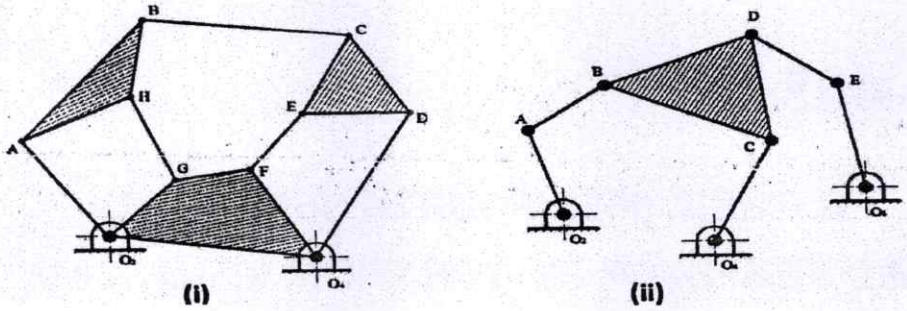
2. (a) Discuss the various types of constrained motion with examples. (CO1, CO2)

OR

(b) Using Grubler's criterion for plane mechanism, prove that the minimum number of binary links in a constrained mechanism with simple hinges is four. (CO1, CO2)

P. T. O.

3. (a) Determine the Degree of Freedom (DoF) of the following mechanisms : (CO1, CO2)



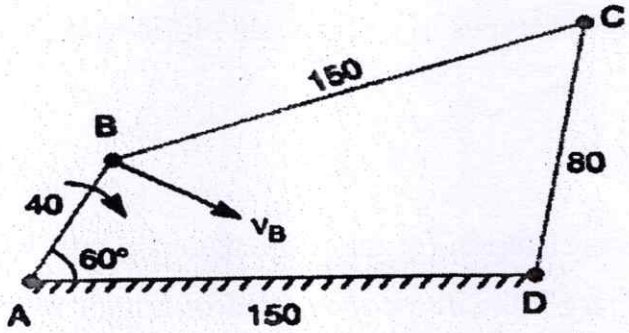
OR

(b) Explain the functions of elliptical Trammel and Oldham's coupling in detail. (CO1, CO2)

4. (a) Describe various inversions of a single slider crank mechanism with examples. (CO2, CO3)

OR

(b) In a four bar chain ABCD as shown fig. below, AD is fixed and is 150 mm long. The crank AB is 40 mm long and rotates at 120 r.p.m. clockwise, while the link CD=80 mm oscillates about D. BC and AD are of equal length. Find the angular velocity of link CD when angle BAD = 60°. (CO2, CO3)



(All dimensions are in mm)

5. (a) Describe the procedure to construct the velocity diagram of a four link mechanism. (CO2, CO3)

OR

(b) In a crank and slotted lever quick return mechanism, the distance between the fixed centres is 150 mm and the driving crank is 75 mm long. Determine the ratio of the time taken on the cutting and return strokes. (CO2, CO3)

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TMS–401

B. TECH. (R & A, AME) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

MANUFACTURING EXECUTION SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write in detail about the description of tomorrow's manufacturing plant. (CO1)

OR

(b) State the differences between Intermittent and Continuous production systems. (CO1)

2. (a) Differentiate between Mass production systems and Cellular production system. (CO1)

OR

(b) Differentiate between short-term, medium-term and long-term planning horizons. (CO1)

3. (a) What are the significances of 'Enterprise Integration' ? Discuss the steps involved for its implementation in an organization. (CO1, CO2)

P. T. O.

OR

(b) Explain in detail about the concept of 'Advanced Manufacturing.' (CO1, CO2)

4. (a) Explain in detail about the concept of 'Material Requirement Planning' along with its primary objectives. (CO2, CO3)

OR

(b) Write in detail about the significance of 'Manufacturing Process Management.' (CO2, CO3)

5. (a) Explain in detail about the concept of 'Manufacturing Execution System' alongwith its benefits. (CO2, CO3)

OR

(b) Explain in detail about the concept of 'Product Lifecycle Management.' Also discuss different phases involved in it. (CO2, CO3)

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TME–403

B. TECH. (MECH. ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

MANUFACTURING PROCESS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Discuss the types of chip formation during machining by single point cutting tool. Explain with the help of sketches. (CO1/CO3)

OR

(b) Discuss functioning of a shaper machine with the help of sketches. (CO1/CO3)

2. (a) Show the complete merchant's circle diagram indicating different forces and angles. (CO1/CO2)

OR

(b) During orthogonal machining of a mild steel specimen with a cutting tool of zero rake angle, the following data is obtained : Cutting Force = 1200 N, Thrust force = 500 N, Rake angle = 0°, Cutting speed = 1 m/s, Uncut chip thickness = 0.25 mm, Chip thickness = 0.75 mm, Width of cut = 2.5 mm. Find the friction and shear angle. (CO1/CO2)

P. T. O.

3. (a) Discuss the *five* cutting tool materials used in marching and also discuss their properties. (CO2/CO3)

OR

(b) Write down the different process performed on a lathe machine. Also, explain them with neat sketches. (CO2/CO3)

4. (a) What is a capstan lathe ? Explain with the help of sketches. (CO3)

OR

(b) Write down the differences between shaper, planer, and slotter machine. (CO3)

5. (a) Explain the process of tapping with neat sketches. (CO3/CO1)

OR

(b) Discuss the modified Taylor's tool life equation. (CO3/CO1)

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TME–404

B. TECH. (ME) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

MEASUREMENT AND INSTRUMENTATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the term SI units and discuss their significance in measurement. (CO1)

OR

(b) Explain the importance of having standardized units in engineering and science. (CO1)

2. (a) Provide examples illustrating situations where accuracy is more critical than precision and vice-versa. (CO2)

OR

(b) Explain the role of instrumentation in the field of measurement. (CO1)

3. (a) How does the quality of measurements impact the reliability of experimental results ? (CO1 & CO2)

OR

(b) Discuss the role of gauges in verifying the dimensions and tolerances of manufactured parts. (CO1 & CO2)

P. T. O.

4. (a) Define systematic and random errors in measurements. (CO1 & CO2)

OR

(b) Describe the characteristics of clearance fit, interference fit and transition fit. (CO1 & CO2)

5. (a) Explain the concept of tolerance. Why is it essential in engineering drawings ? (CO3)

OR

(b) Discuss the purpose of using limit fits in the design and manufacturing of mechanical components. (CO3)

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TEE–404

B. TECH. (EE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

DIGITAL ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Convert methodically the following numbers to decimal numbers : (CO1)

(i) $(6E8F)_{Hex}$

(ii) $(256)_{Octal}$

OR

(b) Write the gray codes of the following decimal numbers : (CO1)

(i) $(15)_{10}$

(ii) $(10)_{10}$

(iii) $(8)_{10}$

(iv) $(0)_{10}$

2. (a) Write the truth table of 'OR', 'AND' and 'NOT' gates. (CO1)

P. T. O.

OR

(b) Write the truth table of universal gate. (CO1)

3. (a) Describe the De Morgan's Theorems. Prove any *one* of the two theorems. (CO1)

OR

(b) Realize the following logic operations using NAND gates. (CO1)

(i) NOT

(ii) AND

(iii) OR

4. (a) Convert the decimal number $(29.125)_{10}$ into its equivalent Binary number. (CO2)

OR

(b) Minimize the four-variable logic function using k-map : (CO2)

$$f(w,x,y,z) = \Sigma m(0, 1, 2, 3, 7, 8, 9, 10, 11, 12, 13)$$

5. (a) Find the number of minterms in the following function : (CO2)

$$f(w, x, y, z) = y' + w'z' + xz'$$

OR

(b) Describe with the help of examples the following : (CO2)

(i) Diminished radix complement

(ii) Radix complement

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TME–407

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

FLUID MECHANICS AND FLUID MACHINES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define fluid. How is it different from a solid ? (CO1)

OR

(b) Explain viscosity of a fluid. (CO1)

2. (a) Derive the hydrostatic relation. (CO1)

OR

(b) On what principle does a U-tube nanometer work ? Explain its working in detail. (CO1)

3. (a) What is stable, unstable, and neutral equilibrium ? How can a boat be made to be stable while floating ? (CO1, CO2)

OR

(b) Derive continuity equation for flow in a pipe. (CO1, CO2)

P. T. O.

4. (a) What is velocity potential ? Why is it useful ? (CO2)

OR

(b) For $u = \left(5x, 4y^2\right)$ and $v = \left(3x^3, 8y\right)$. Is this flow field physical or not ? Why ? (CO2)

5. (a) Derive Euler's equation of motion with diagram. (CO2)

OR

(b) What are the *three* heads of Bernoulli equation ? (CO2)

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TME–408

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

APPLIED THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write the various assumptions made in the analysis of air standard cycle. (CO1)

OR

(b) Define the various performance parameters of an internal combustion engine. (CO1)

2. (a) The minimum pressure and temperature in an Otto cycle are 150 kPa and 300 K. The amount of heat added to the air per cycle is 2000 kJ/kg. (CO3, CO6)

(i) Determine the pressures and temperatures at all points of the air standard Otto cycle.

(ii) Also calculate the workdone per kg and thermal efficiency of the cycle for a compression ratio of 8 : 1.

P. T. O.

OR

(b) The pressure and temperature of air at the beginning of compression in an Otto cycle is 120 kPa and 330 K, respectively. The heat added per kg of air is 1850 kJ. The compression ratio is 8. Determine : (CO3, CO6)

(i) maximum temperature,

(ii) maximum pressure, and

(iii) thermal efficiency.

3. (a) Write the name of different processes constitute the following air standard cycles and also draw their P-v and T-s diagram : (CO2)

(i) Erikson's cycle

(ii) Lenoir's cycle

(iii) Atkinson's cycle

OR

(b) Write the principle of jet and rocket propulsion. Also, explain the working of pulse jet and turbo jet engine with the help of a neat sketch. (CO2, CO4)

4. (a) Explain the air standard Brayton cycle with the help of P-v and T-s diagram and also prove that its efficiency depends on pressure ratio only. (CO3)

OR

(b) Air enters the compressor of a gas turbine plant operating on Brayton cycle at 101.325 kPa, 27°C. The pressure ratio in the cycle is 6. Assume

W_T = 2.5 times of W_C, where W_T and W_C are the turbine and the compressor work respectively. Calculate : (CO3, CO6)

(i) the maximum temperature in the cycle,

(ii) net workdone, and

(iii) the cycle efficiency.

5. (a) Explain the effect of regeneration and intercooling on Brayton cycle with the help of T-s diagram. (CO4)

OR

(b) The pressure ratio and maximum pressure of an air standard Brayton cycle are 5 : 1 and 923 K, respectively. Air enters the compressor at 1 bar and 298 K, calculate for 1 kg of air flow : (CO3, CO6)

(i) the compressor work,

(ii) turbine work, and

(iii) efficiency of cycle.

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